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# Applying Named Entity Extraction to the Archelec Corpus: An Analysis of Candidate Speeches

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## Abstract

This paper presents a natural language processing pipeline applied to the Archelec corpus, a large collection of French electoral manifestos spanning legislative elections from 1973 to 1993. We evaluate three named entity recognition approaches, SpaCy, GLiNER, and CamemBERT, in both zero-shot and supervised settings. While CamemBERT achieves the highest in-domain performance ( $F1 = 46.3\%$ ), GLiNER fine-tuned is selected for large-scale inference owing to its better generalization and scalability. Applied to over 8,000 documents, GLiNER is used to extract presidential mentions, which are subsequently classified as positive or negative using a TF-IDF + XGBoost pipeline trained on 200 manually annotated excerpts. Results reveal that references to sitting presidents in electoral manifestos are overwhelmingly positive, with variations reflecting party affiliation, political context, and presidential era.

## 1 Introduction

Named Entity Recognition (NER) is a fundamental task in natural language processing, enabling the automatic extraction of structured information from large volumes of unstructured text. In the context of historical political corpora, NER offers a powerful means to trace how candidates frame their discourse, which figures they invoke, and how they situate themselves geographically and politically.

In this study, we apply a range of NER approaches to the Archelec corpus, a large collection of French electoral manifestos (*professions de foi*) covering legislative elections from 1973 to 1993. We evaluate methods spanning from lightweight, off-the-shelf tools such as SpaCy to more advanced transformer-based models fine-tuned on a small manually annotated dataset. The comparison is conducted in both zero-shot and supervised settings, with a particular focus on generalization across time periods.

Beyond the NER evaluation, we leverage the best-performing model to extract mentions of national political leaders at scale, and analyse how local candidates rhetorically position themselves with respect to the sitting president. To this end, we develop a sentiment classification pipeline to determine whether presidential references reflect a stance of alignment or opposition, and examine how these patterns vary across parties, political blocs, and presidential eras.

## 2 State-of-the-art

Before beginning our modeling work, we explored existing Named Entity Recognition techniques. This field has undergone several major paradigm shifts, evolving from rule-based systems to large language models. The detection and evaluation framework used today was introduced by Grishman and Sundheim (1995) in MUC-6[3]. They defined entity types (PERSON, LOCATION, MONEY, ORGANIZATION, DATE, TIME, PERCENT) and established evaluation metrics still commonly used today : recall, precision, and F1-score to evaluate rule-based methods. State-of-the-art techniques

at the time were rule-based and domain-dependent. One example is FASTUS (Finite State Automaton Text Understanding System) introduced by Hobbs et al. (1997)[4]. FASTUS processes text through five cascaded finite-state stages : Stage 1 recognizes named entities and multi-word expressions ; Stage 2 segments sentences into basic syntactic units (noun groups, verb groups, particles) ; Stage 3 builds complex phrases and captures semantic equivalences ; Stage 4 applies domain-specific patterns to extract events of interest, automatically generating syntactic variants from user-defined templates ; Stage 5 merges coreferent structures across the document to produce unified database entries.

While these early methods showed good performance, they required significant manual effort to define patterns for each new domain and struggled to capture statistical regularities in language. Consequently, the late 1990s and early 2000s marked a transition to statistical and data-driven methods. New approaches emerged, such as maximum entropy introduced by Borthwick (1999) [1], the first statistical NER system to demonstrate that probabilistic models could match or exceed rule-based performance while offering superior portability across domains and languages. The system uses maximum entropy modeling to combine diverse knowledge sources (lexical features, dictionaries, external NER system outputs, and reference resolution) into a unified probabilistic framework that learns patterns automatically from annotated data. This was followed by Lafferty, McCallum, and Pereira (2001) [5], who introduced Conditional Random Fields (CRFs), a discriminative probabilistic framework for sequence labeling that models the conditional probability  $P(\text{labelsequence}|\text{observedsequence})$  using a log-linear combination of arbitrary feature functions. This enabled the integration of rich, overlapping features such as word context, capitalization, prefixes, suffixes, and gazetteers without requiring independence assumptions. This approach achieved state-of-the-art performance that remained unchallenged until the deep learning revolution.

Deep learning architectures (CNNs and LSTMs) were applied to NER tasks in the mid-2010s. Lample et al. (2016) introduced the BiLSTM-CRF architecture [6], pioneering representation learning by combining character-level representations (via LSTMs) with bidirectional word-level LSTMs and a CRF output layer for joint sequence decoding. The model achieved state-of-the-art results without any hand-crafted features, gazetteers, or language-specific resources, demonstrating that end-to-end neural architectures could learn optimal representations directly from data while being more portable across languages than feature-engineering approaches.

However, static word embeddings could not capture polysemy, context dependent representations were needed. This led to the transformer and pre-training era (2018-2022), where Devlin et al. (2019) introduced BERT [2], revolutionizing NER and all NLP tasks through bidirectional transformer pre-training. BERT’s contextualized representations enabled simple fine-tuning to achieve state-of-the-art NER performance, making feature engineering largely obsolete. Language-specific models like CamemBERT demonstrated that pre-training on monolingual corpora could outperform multilingual models for specific languages, advancing NER performance for French.

Beyond research systems, spaCy emerged as a leading production ready NER library, providing pre-trained models that evolved from combining CNN and BiLSTM architectures to incorporating transformer based models, with an industrial focus on speed and accuracy.

More recently, the paradigm shifted toward zero-shot and few-shot learning with large language models. Zaratiana et al. (2023) introduced GLiNER (Generalist Model for Named Entity Recognition) [7], a generalist model achieving zero-shot NER across arbitrary entity types without fine-tuning. Unlike traditional approaches requiring task-specific training data, GLiNER can recognize new entity types on-the-fly simply by specifying them at inference time, demonstrating that instruction following capabilities could eliminate the need for domain specific fine-tuning in many applications.

### 3 Named-Entity Extraction

In order to identify named entities, we adopted three different approaches, combining both zero-shot (without training) and supervised (with training) settings : SpaCy, CamemBERT, and GLiNER. To determine the most effective approach, we rely on the global F1 score as our primary evaluation metric. In addition, we conducted a fine-grained analysis by examining the per-entity-type F1 score, so as to assess the strengths and limitations of each method across the different entity categories.

### 3.1 Dataset

All available files from the Archelec corpus were retrieved from the official repository<sup>1</sup>. Recent files (from 2015 onwards) were added to evaluate model performance on more contemporary data. Older files (prior to 2000), which amount to 10 598 documents, will be used in their entirety in the analysis presented in Part 2 of this report.

**Data annotation :** For the supervised approach, we used a subset of the Archelec corpus. The annotation process was carried out in two successive phases.

The first phase combined metadata matching with rule-based regex patterns. This phase successfully identified mentions of political parties, substitute candidates, main candidates, as well as their respective professions and geographical information (departments and constituencies). However, this approach proved insufficient for detecting mentions of competing parties, references to heads of state, foreign countries, and other regions not directly covered by the metadata.

To address these limitations, a second phase of manual annotation was conducted on a subset of the data, covering both the training and test sets used in the supervised setting. The details of this annotation are as follows :

**Training Set :** The training set consists of 10 political manifestos per year, covering the legislative elections held between 1973 and 1993 (inclusive).

**Test Set :** The evaluation was conducted in two distinct phases :

- **Phase 1 — In-domain evaluation :** 10 political manifestos per year, drawn from the same legislative elections period spanning **1973 to 1993** (inclusive), mirroring the temporal scope of the training set.
- **Phase 2 — Out-of-domain evaluation :** 10 political manifestos per election type and per year, covering the following elections :European Elections (2019), Legislative Elections, (2017), Regional Elections (2015), Legislative Elections (2020), Municipal Elections (2020)

### 3.2 SpaCy

The SpaCy-based experiments were conducted in two settings : zero-shot and supervised.

In the zero-shot setting, the pre-trained French model `fr_core_news_sm` was used directly for inference, without any fine-tuning on the Archelec data. For each document, the model processes the raw text and extracts named entity spans, each characterized by its surface form, label, and character offsets.

In the supervised setting, a blank French SpaCy pipeline was initialized and fine-tuned on the annotated training set. The training data was converted into SpaCy’s native binary format (`DocBin`), with entity spans aligned to character offsets using the `contract` alignment mode to handle tokenization mismatches. The model was trained for 10 epochs using the built-in NER pipeline, with a learning rate of  $10^{-3}$ , targeting four entity categories : PER, ORG, LOC, and MISC. The best model checkpoint was retained for evaluation.

In both settings, inference was performed separately on the two test splits (before and after 2000), and the resulting predictions were evaluated using a custom evaluation function computing precision, recall, and F1 score per entity type as well as globally.

### 3.3 CamemBert

CamemBERT is a masked language model of the RoBERTa family, pre-trained on 138 GB of French text, which we adapt to NER via token classification using a linear BIO tagging head covering 9 labels (B/I-PER, B/I-ORG, B/I-LOC, B/I-MISC, and O). To compensate for the limited amount of manually annotated data, we adopt a two stage curriculum learning strategy. In the first stage, the model is pre-trained on 6,936 automatically annotated documents using metadata, for 5 epochs with a learning rate of  $2 \times 10^{-5}$  and early stopping on global F1. This step serves as domain adaptation, allowing the model to internalize the vocabulary and style of French political manifestos before exposure to any gold labels. In the second stage, the resulting domain-adapted model is fine-tuned on the 50 manually annotated documents with a reduced learning rate of  $1 \times 10^{-5}$  and a patience-based early stopping

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1. [https://gitlab.teklia.com/ckermorvant/arkindex\\_archelec](https://gitlab.teklia.com/ckermorvant/arkindex_archelec)

over up to 20 epochs. Validation is performed on 30 documents from the 1973-1993 period, while the final evaluation uses 50 test documents held out throughout the entire training process.

### 3.4 Gliner

GLiNER is a span-based model that performs zero-shot NER by matching text spans against natural language entity labels, without requiring task-specific training data. The experiments were conducted in two settings : zero-shot and fine-tuned.

In the zero-shot setting, the multilingual pre-trained model `urchade/gliner_multi-v2.1` was used directly for inference. Since GLiNER operates on token sequences of bounded length, long documents were split into overlapping chunks of at most 300 characters, with cuts performed at sentence boundaries (periods, exclamation marks, question marks, or newlines) to preserve linguistic coherence. Entity labels were expressed as natural language descriptors : *person*, *location*, *political party/political movement*, and *profession*, and subsequently mapped to the standard NER tags PER, LOC, ORG, and MISC. A confidence threshold was swept across values from 0.1 to 0.9 to determine the optimal operating point, which was found to be 0.8.

In the fine-tuned setting, the same base model was adapted on the annotated training set. Documents were tokenized using GLiNER’s internal whitespace-based tokenizer and split into chunks of at most 384 tokens, with entity spans converted to chunk-relative, inclusive token indices following GLiNER’s training convention. To limit computational cost while preserving representational capacity, the transformer backbone (DeBERTa) was largely frozen : only the last two encoder layers (layers 10 and 11, out of 12) were kept trainable, resulting in approximately 22 million trainable parameters out of 195 million total. The model was trained for 800 steps with a learning rate of  $10^{-4}$ , a linear scheduler with 20 warm-up steps, and a focal loss ( $\alpha = 0.75$ ,  $\gamma = 2$ ) to handle class imbalance. Evaluation was then performed across the same threshold range as in the zero-shot setting.

### 3.5 Results and discussion

Table 1 reports exact match F1 scores on the before 2000 test set across all models and entity types. Fine-tuning consistently improves over the zero-shot setting for all models : SpaCy gains 12.5 points (11.8  $\rightarrow$  24.3), and GLiNER gains 11.3 points (23.7  $\rightarrow$  35.0). CamemBERT achieves the best overall performance (46.3%), benefiting from its two-stage curriculum strategy and its deeper contextual representations.

At the per-type level, PER is the best-detected entity across all models and settings, which can be explained by the predictable and structured surface forms that person names typically take in political manifestos, most candidates are referred to by their first name and last name, making them easier to identify regardless of the model or training regime.

LOC is well captured by fine-tuned models, particularly CamemBERT (55.2%) and SpaCy (41.3%), whereas zero-shot models struggle with this category. Interestingly, GLiNER fine-tuned reaches an F1 of 45% on LOC at a higher confidence threshold of 0.9, suggesting that the model does identify location entities reliably, but tends to produce lower-confidence predictions that are filtered out at stricter thresholds.

ORG remains moderately difficult across the board, likely due to the high variability of political party names and abbreviations across decades.

MISC, used here to label professions, is the most challenging category : it reaches 0.0 for both SpaCy variants and CamemBERT, and only GLiNER fine-tuned achieves a non-negligible score (18.8%), which can be attributed to its use of natural language label descriptors during inference.

Table 2 reveals a stark contrast in generalization ability between zero-shot and fine-tuned models. Zero-shot models, having never been exposed to the training distribution, are by construction unbiased towards the before 2000 period and maintain stable, or even slightly improved performance on post 2000 documents : SpaCy zero-shot gains 3.8 points and GLiNER zero-shot remains virtually unchanged (+0.6). Fine-tuned models, on the other hand, exhibit severe performance drops when evaluated on post-2000 data. SpaCy fine-tuned collapses from 24.3 to 2.5 (−21.8 points), and CamemBERT suffers the largest degradation (−27.4 points, from 46.3 to 18.9). This can be attributed to the significant shift in political discourse between the two periods : the vocabulary, the named

TABLE 1 – NER results, exact match F1 scores (%) per model and entity type. All scores reported on the before-2000 test set. GLiNER results use a confidence threshold of 0.8. CamemBERT is fine-tuned.

| Model                  | Global F1 | PER   | ORG   | LOC   | MISC |
|------------------------|-----------|-------|-------|-------|------|
| SpaCy (zero-shot)      | 11.8      | 30.4  | 13.8  | 9.2   | 1.6  |
| SpaCy (fine-tuned)     | 24.3      | 52.1  | 17.8  | 41.3  | 0.0  |
| GLiNER (zero-shot)     | 23.7      | 52.1  | 15.9  | 15.3  | 5.1  |
| GLiNER (fine-tuned)    | 35.0      | 66.7  | 21.5  | 39.8  | 18.8 |
| CamemBERT (fine-tuned) | 46.3      | 73.95 | 36.02 | 55.17 | 0.0  |

TABLE 2 – Generalization analysis, exact match global F1 (%) before and after 2000. A large drop between the two splits indicates poor out-of-domain generalization.

| Model                  | F1 before 2000 | F1 after 2000 | Drop  |
|------------------------|----------------|---------------|-------|
| SpaCy (zero-shot)      | 11.8           | 15.6          | +3.8  |
| SpaCy (fine-tuned)     | 24.3           | 2.5           | −21.8 |
| GLiNER (zero-shot)     | 23.7           | 24.3          | +0.6  |
| GLiNER (fine-tuned)    | 35.0           | 26.8          | −8.2  |
| CamemBERT (fine-tuned) | 46.3           | 18.9          | −27.4 |

entities referenced, and the rhetorical style of manifestos evolved substantially from the 1970s–1990s to the 2010s–2020s, introducing named entities, new party names, emerging political figures, and international references, that were absent from the training data. GLiNER fine-tuned proves the most resilient among supervised models, with a more moderate drop of 8.2 points (35.0 → 26.8), reflecting the greater robustness of its span-based, label-conditioned architecture to distributional shift.

In light of these results, we select GLiNER fine-tuned for the large-scale annotation of the full Archelec corpus. Despite CamemBERT’s higher in-domain F1, GLiNER offers a more robust and scalable solution : its inference pipeline does not require domain-specific retraining, and its generalization gap is better across time periods. An additional advantage of GLiNER is the flexibility of its confidence threshold, which can be tuned post-inference independently for each entity type. This property is particularly valuable in our setting, as the optimal threshold differs between PER and LOC entitie, a single threshold would inevitably favour one category at the expense of the other. In the remainder of this report, GLiNER fine-tuned will be applied to detect person names, specifically national political leaders, and locations across the entire corpus, with a focus on documents predating 2000, in order to support a broader analysis of political discourse and its geographical and personal references over time.

## 4 Data analysis

In the following section, we apply the fine-tuned GLiNER model to extract named entities at scale across the pre-2000 corpus, with the aim of deriving insights from the political discourse of local candidates. We focus on one central research question : how local candidates rhetorically associate themselves with national political leaders in their manifestos, and whether these references reflect a strategy of alignment or distancing with respect to the incumbent president.

### 4.1 Does the candidates link themselves positively or negatively to national leaders ?

### 4.2 Data pre-processing and modelisation

We applied GLiNER fine-tuned to the entire pre-2000 corpus to extract person entities. Each predicted entity was then matched against the name of the sitting president at the time of the election : Pompidou for 1973, Valéry Giscard d’Estaing for 1978, and François Mitterrand for the 1981–1993 period. Mentions that did not match any of these names were discarded. From the remaining matches,

we constructed a dataset by retaining the surrounding context of each mention, consisting of 300 characters before and after the detected name, so as to capture the rhetorical framing used by candidates when referring to the incumbent president.

To analyse the nature of these references, we classified 200 randomly selected excerpts using a classification pipeline built on the Anthropic API, and manually verified the outputs to assess their reliability.

Each excerpt was classified into one of two categories. A mention was labelled as *positive* when the candidate explicitly supports, endorses, or associates themselves with the incumbent president, for instance by invoking the president’s name to legitimise their own candidacy, align their programme with the government’s action, or express political loyalty. Conversely, a mention was labelled as *negative* when the candidate distances themselves from or criticises the president, whether by rejecting the government’s record, using the president’s name as a counter reference, or framing their candidacy as an opposition to the incumbent’s policies.

This initial classification step was designed as a cost efficient pilot : by validating a small sample first, we aimed to confirm the pipeline’s accuracy before scaling it up to classify the remaining 4,467 excerpts in the full dataset.

We designed two classification pipelines and evaluated them through 5-fold stratified cross-validation to ensure robust estimates despite the small dataset size.

The first pipeline combines a TF-IDF representation (10,000 features, unigrams and bigrams, `sublinear_tf=True`, French stopwords removed) with an XGBoost classifier. Class imbalance was addressed jointly by applying SMOTE oversampling within each training fold and by setting XGBoost’s `scale_pos_weight` parameter proportionally to the class ratio. The second pipeline replaces the TF-IDF representation with contextual embeddings extracted from `camembert-base` : the [CLS] vector (768 dimensions, truncated to 256 tokens) is used as a fixed-length document representation, which is then fed into a CatBoost classifier, with class imbalance handled through SMOTE on the embedding space combined with fold-adaptive class weights.

As shown in Figure 1, TF-IDF + XGBoost consistently matches or outperforms CamemBERT + CatBoost across all metrics, achieving a higher macro F1 (0.64 vs 0.62) and a notably better ROC-AUC (0.82 vs 0.71). Both pipelines struggle with the negative class (F1  $\approx$  0.30), which is expected given the class imbalance. The TF-IDF + XGBoost pipeline was therefore selected for final inference on the 6,142 unannotated excerpts, as it achieves better discriminative performance while remaining significantly lighter than the CamemBERT-based alternative.

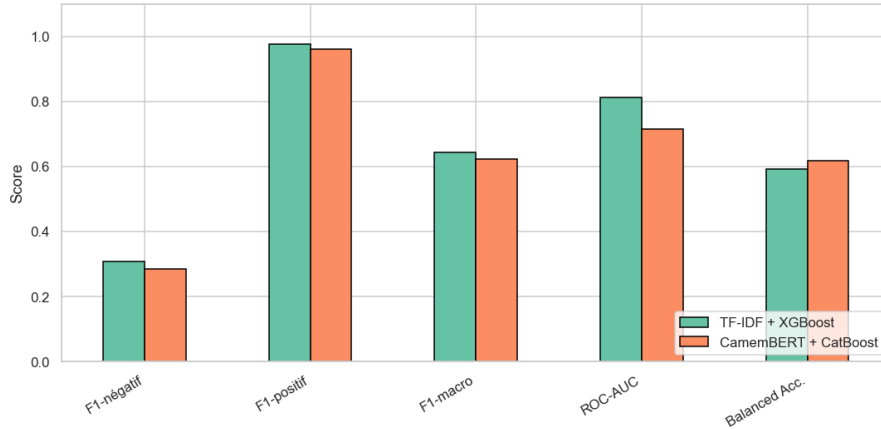


FIGURE 1 – Comparison of the two classification pipelines across five evaluation metrics (5-fold stratified cross-validation).

### 4.3 Results

#### Interpretability : Discriminative Keywords

Before examining the predicted sentiments at scale, we first analyse the most discriminative features learned by the TF-IDF + XGBoost classifier. Among the features most positively associated with a positive sentiment, the terms *alliance*, *élu* and *majorité* stand out. These words reflect a rhetoric of political alignment : candidates who invoke the president tend to frame him as a unifying figure, a legitimate elected leader, or the architect of a governing majority. Conversely, the terms most negatively associated with positive sentiment, *voter*, *faut*, *changer*, *commun*, evoke a vocabulary of urgency and rupture, typical of opposition discourse calling for electoral change or rejecting the incumbent's record. This lexical contrast is politically coherent and suggests that the classifier has captured genuine rhetorical signals rather than spurious correlations.

### Overall Trends : Proportion of Positive Mentions by Electoral Period

Figure 2 displays the proportion of positive presidential mentions per electoral year. A first observation is that all data points lie well above the 50% threshold, indicating that references to the sitting president in political manifestos are globally positive across the entire period. This result is in itself noteworthy : even opposition candidates avoid mentioning the president or tend to mention him in relatively neutral or contextualised terms, rather than systematically attacking him.

Pompidou (1973) records the highest proportion of positive mentions (95%,  $n = 318$ ), which is consistent with the political landscape of the early 1970s : the Gaullist bloc was dominant, and most parties that mentioned the president were aligned with or close to his majority. The sharp decline observed between 1973 and 1988 traces the progressive erosion of presidential consensus as the political landscape became more competitive and polarised. Giscard d'Estaing (1978, 73%,  $n = 131$ ) and Mitterrand's first term (1981, 69%,  $n = 1966$ ) show a continued downward trend. It is worth noting that the 1981 figure is based on a substantially larger sample ( $n = 1966$ ), making it the most statistically reliable estimate in the series.

François Mitterrand's trajectory across his two terms is particularly informative. His first term (1981, 69%) is characterised by a relatively moderate positivity rate, reflecting a mixed reception : while left-wing parties strongly supported him, centrist and right-wing candidates were far more critical. By his second term (1988, 64%,  $n = 1011$ ), positivity reaches its lowest point in the entire series, despite cohabitation having ended. This period coincided with growing disillusionment on both sides of the political spectrum, with the left increasingly disappointed by the government's economic record and the right reluctant to endorse him. The recovery observed in 1993 (83%,  $n = 95$ ) should be interpreted with caution : the much smaller sample size ( $n = 95$ ) limits its statistical weight, and the high positivity likely reflects a selection effect, as fewer parties mentioned Mitterrand by name during this late period of his presidency.

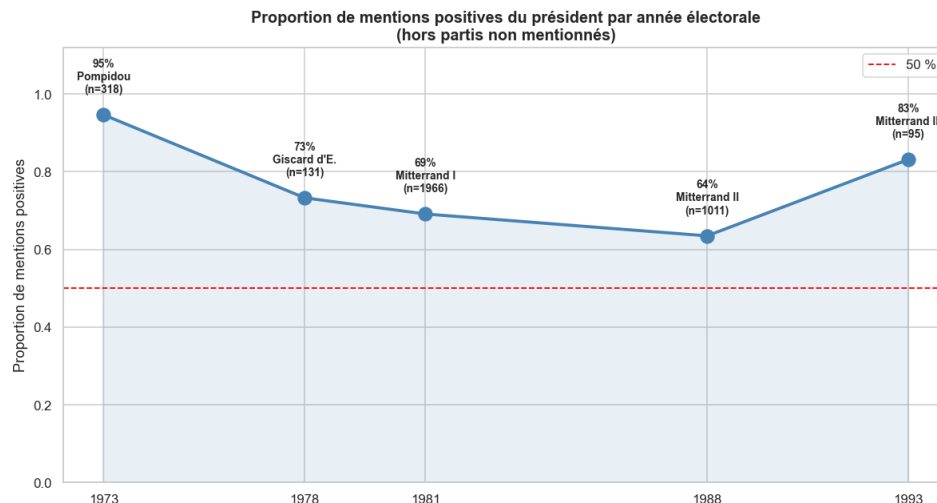


FIGURE 2 – Proportion of positive presidential mentions per electoral year (parties with no declared affiliation excluded).

### Political Bloc Analysis : Sentiment by Party Orientation

Figure 3 disaggregates positive mention rates by presidential era and political bloc. Three blocs were defined based on party affiliation : *Droite / Centre* (including Gaullist, republican, centrist, and radical parties such as RPR, UDF, and affiliated movements), *Extrême droite* (Front National exclusively), and *Gauche* (socialist, communist, and far-left parties including the Parti socialiste, Parti communiste français, Lutte ouvrière, and Parti socialiste unifié).

The heatmap reveals clear and politically interpretable patterns. For Pompidou (1973), the *Droite / Centre* bloc shows a near-unanimous positivity rate of 99%, which is expected given that Pompidou was himself a Gaullist. Strikingly, the *Extrême droite* bloc records only 10% positive mentions, reflecting the Front National’s adversarial posture toward the Gaullist establishment from its very early years. The *Gauche* bloc reaches 67%, suggesting that even left-wing candidates occasionally referenced Pompidou in relatively neutral terms, possibly to contextualise their opposition.

For Giscard d’Estaing (1978), the *Droite / Centre* bloc remains strongly positive (83%), while the *Gauche* bloc drops to 62%, a moderate but meaningful decline consistent with the left’s growing electoral ambitions under a united programme (*Programme commun*). The absence of *Extrême droite* data for this period reflects the Front National’s marginal electoral presence in 1978.

Mitterrand’s two terms present the most analytically rich pattern. During his first term (1981), the *Gauche* bloc records 68% positivity — lower than might be expected for a left-wing president supported by left-wing candidates. This suggests that even within his own camp, Mitterrand was referenced in a variety of rhetorical contexts, not exclusively celebratory. The *Droite / Centre* bloc reaches 88%, which may seem counterintuitive but reflects the fact that right-wing candidates often mentioned Mitterrand to frame their opposition, using his name as a reference point rather than as an endorsement. During his second term (1988), the *Extrême droite* bloc reaches its peak at 94%, a paradoxical figure that likely reflects the Front National’s tactical use of Mitterrand’s name in a context of cohabitation, positioning themselves as an alternative to both the left and the mainstream right.

The most striking observation concerns Mitterrand’s third electoral period (1993) : the *Droite / Centre* bloc collapses to 0% positivity, a clear signal of the hostility of the right-wing opposition at the end of his second term, marked by economic difficulties and political scandals. Conversely, the *Gauche* bloc rises to 86%, as left-wing parties rallied around a besieged president in a defensive posture.

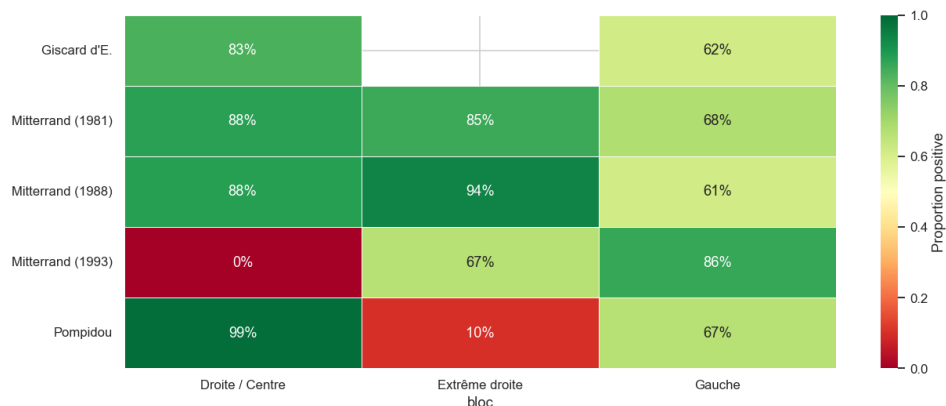


FIGURE 3 – Proportion of positive presidential mentions by political bloc and presidential era. The three blocs are defined as follows : *Droite / Centre* (Gaullist, republican, and centrist parties), *Extrême droite* (Front National), and *Gauche* (socialist, communist, and far-left parties). Empty cells indicate an absence of mentions for that bloc in the corresponding period.

### Party-Level Analysis : Top 5 Parties by Electoral Year

Figure 4 provides a finer-grained view by identifying the five parties with the highest number of presidential mentions per electoral year, along with their positivity rate.

Across all years combined, the Parti socialiste dominates with 1,506 mentions (69% positive), followed by the Parti communiste français (950 mentions, 59% positive). These two parties account for the vast majority of presidential references in the corpus, which is consistent with the fact that

Mitterrand, a socialist president, held office for most of the period covered (1981–1993). The Union des républicains de progrès stands out with 201 mentions and a near-perfect positivity rate (99%), reflecting the cohesion of the Gaullist bloc around Pompidou in 1973. The Parti socialiste unifié (92% positive,  $n = 96$ ) and the Union pour la démocratie française (72%,  $n = 116$ ) complete the overall top 5.

In 1973, presidential mentions are concentrated among Gaullist and centre-right parties : the Union des républicains de progrès (201 mentions, 99% positive) and the Alliance républicaine indépendante et libérale (65 mentions, 98% positive) overwhelmingly support Pompidou, while the Front national (10 mentions, 10% positive) and centrist parties show more reserved or hostile stances. The small sample sizes for opposition parties in 1973 limit the interpretability of their percentages.

In 1978, the Union pour la démocratie française leads with 61 mentions (67% positive), followed by the Parti républicain (33 mentions, 82% positive), both aligned with Giscard d'Estaing. The Rassemblement pour la République (7 mentions, 100% positive) is notable : the RPR, a Gaullist party, supported Giscard despite internal rivalries, at least in the framing of their manifestos. The small overall mention count in 1978 reflects the limited penetration of presidential references in that election cycle.

The 1981 and 1988 elections generate by far the largest volumes of mentions, making them the most statistically robust periods for analysis. In 1981, the Parti socialiste alone accounts for 1,146 mentions (69% positive), suggesting that socialist candidates referenced Mitterrand frequently but not uniformly positively, consistent with an internal diversity of positioning. The Parti communiste français (421 mentions, 59% positive) shows a more reserved tone, reflecting the historically tense relationship between the PS and the PCF despite their electoral alliance. In 1988, the Parti communiste français leads with 512 mentions (58% positive) and the Parti socialiste follows with 332 mentions (67% positive). Remarkably, the Front national records 32 mentions of Mitterrand with a 94% positivity rate, suggesting a tactical appropriation of the president's name. By 1993, mention volumes collapse sharply across all parties, with the Parti socialiste retaining only 25 mentions (84% positive), reflecting Mitterrand's diminished political centrality in the final years of his presidency.

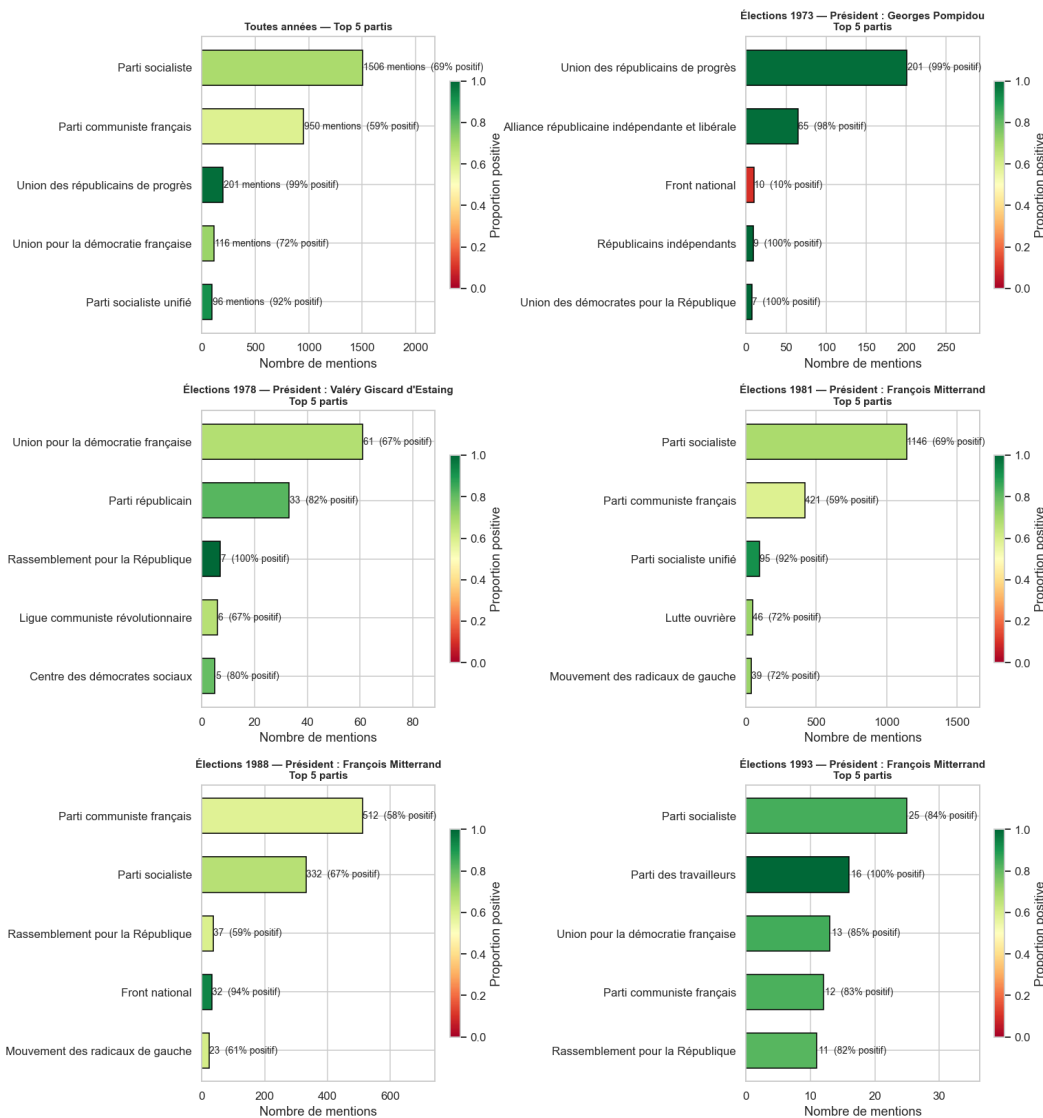


FIGURE 4 – Top 5 parties by number of presidential mentions per electoral year (and across all years combined).

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